

CV - Matthew Francisco

Director of Undergraduate Studies, Department of Informatics
Senior Lecturer, Luddy School of Informatics, Computing, and Engineering
Indiana University Bloomington
Bloomington, Indiana | francm@indiana.edu

1. CURRENT AND PAST EMPLOYMENT

Director of Undergraduate Studies, Department of Informatics
Indiana University Bloomington, July 2025 - Present

Senior Lecturer, Luddy School of Informatics, Computing, and Engineering
Indiana University Bloomington, July 2022 - Present

Lead Instructor and Director of Curriculum, Luddy Precollege Summer Program
Indiana University Bloomington, January 2021 - August 2024

Lecturer, Luddy School of Informatics, Computing, and Engineering
Indiana University Bloomington, 2016 - 2022

Visiting Lecturer, School of Informatics and Computing
Indiana University Bloomington, 2014 - 2016

Research Associate, School of Informatics and Computing
Indiana University Bloomington, 2013 - 2015

Systems Architect, School of Informatics and Computing
Indiana University Bloomington, 2012 - 2014

Visiting Research Associate, School of Informatics and Computing
Indiana University Bloomington, 2010 - 2012

Team Facilitator, Emerging Minority Business Leaders Institute
Wheeling Jesuit University and West Liberty University, 2006, 2008, 2010, and 2011

Teaching Assistant, Department of Science and Technology Studies
Rensselaer Polytechnic Institute, 2004, 2007-2008

Graduate Research Assistant, Department of Computer Science
Rensselaer Polytechnic Institute, 2004-2007

2. EDUCATION

MS/PhD, Science and Technology Studies (August 2010)

Rensselaer Polytechnic Institute, Troy, New York

Advisor: Dr. Linnda Caporael

Dissertation: *Agents on the Loose: Embodied Reflexive Practice in Emerging Computational Social Science*

Certificate, Technology Transfer and Commercialization (August 2002)

Wheeling Jesuit University and the National Technology Transfer Center, Wheeling, West Virginia

BA, Southwest Studies (December 2001)

Fort Lewis College, Durango, Colorado

Minors: Computer Science; Music

3. MOST SIGNIFICANT ACHIEVEMENTS SINCE JANUARY 2019

- **Promoted to Senior Lecturer** (July 2022), recognizing sustained excellence in teaching and service
- **Led Luddy Precollege Summer Program** as Director of Curriculum (2021-2024), designing and implementing innovative STEM education programs for high school students
- **Received Test of Time Award** from International AAAI Conference on Web and Social Media (2021) for influential 2011 research on political polarization on Twitter

4. RESEARCH AND TEACHING INTERESTS

My research focuses on the intersection of technology, society, and education, employing interdisciplinary approaches that combine science and technology studies with computational methods to understand how technological systems shape and are shaped by social practices. Current research areas include computational social science, with particular emphasis on agent-based modeling of classrooms and learning systems; AI-assisted problem discovery frameworks that scaffold student research planning and critical thinking; and technology ethics pedagogy that addresses contemporary challenges in responsible AI education. My work on human-centered robotics and physical computing explores how hands-on prototyping develops students' systems thinking while fostering critical reflection on technology's societal implications.

In teaching, I specialize in computer ethics, physical computing, and educational technology, with a commitment to making complex technical concepts accessible through problem-based learning experiences. My pedagogical approach emphasizes authentic problem-solving that combines practical computing skills with critical analysis of technology's role in society, preparing students to be both technically competent and ethically-informed practitioners.

5. FELLOWSHIPS AND AWARDS

- International AAAI Conference on Web and Social Media (ICWSM) Test of Time Award 2021
- National Science Foundation Dissertation Improvement Award, 2008
- Rensselaer Graduate Fellowship, 2003-2004
- Outstanding Student in Entrepreneurial Technology Apprenticeship Program, 2002
- Fort Lewis College Dean's List, 1999-2001
- Chief Manuelito Academic Excellence Award, 1997-2001

6. GRANTS

Senior Personnel

NSF Innovative Technology Experiences for Students and Teachers (January 2015): "Human-Centered Robotics Experiences for Exploring Engineering, Computer Science, and Society." PIs: Cindy Hmelo-Silver (IU), Selma Sabanovic (IU), Seta Bogosyan (UAF), Orion Lawlor (UAF)

Senior Personnel

Indiana University Collaborative Research Grant (March 2013): "Social SLAM: Creating Dynamical Socio-Environmental Models for Mobile Robots." PIs: Selma Sabanovic and Luis Rocha

Co-Principal Investigator

National Science Foundation Dissertation Improvement Grant (January 2008): "Doctoral dissertation research: Impact of agent-based computational modeling on the sociological imagination of an emerging academic field." PI: Linnda Caporael

7. PUBLICATIONS

Journal Articles

Francisco, M., & Wu, Y. (2025). Computer and Information Ethics in the Era of AI: A Pedagogical Design for Conceptual Muddles. *Proceedings of the 7th Annual Symposium on HCI Education*, 1-10.

Gomoll, A., Šabanović, S., Tolar, E., Hmelo-Silver, C. E., Francisco, M., & Lawlor, O. (2018). Between the Social and the Technical: Negotiation of Human-Centered Robotics Design in a Middle School Classroom. *International Journal of Social Robotics*, 10(3), 309-324.

Gomoll, A., Hmelo-Silver, C., Tolar, E., Šabanović, S., & Francisco, M. (2017). Moving Apart and Coming Together: Discourse, Engagement, and Deep Learning. *Journal of Educational Technology & Society*, 20(4), 219-232.

Gomoll, A., Hmelo-Silver, C. E., Šabanović, S., & Francisco, M. (2016). Dragons, Ladybugs, and Softballs: Girls' STEM Engagement with Human-Centered Robotics. *Journal of Science Education and Technology*, 1-16.

Francisco, M., Milojevic, S., and Sabanovic, S. (2011). Conference models to bridge micro and macro studies of science. *Journal of Artificial Societies and Social Simulation*, 14(4), 13.

Baumes J., Chen H., Francisco M., Goldberg M., Magdon-Ismail M., and Wallace W. (2008). ViSAGE: A virtual laboratory for the simulation and analysis of social group evolution. *ACM Transactions on Autonomous and Adaptive Systems*.

Conference Papers and Presentations

M. D. Conover, B. Gonçalves, J. Ratkiewicz, M. Francisco, A. Flammini, and F. Menczer (2011). Political polarization on twitter. *Proceedings of the 5th International Conference on Weblogs and Social Media*.

Francisco, M., Wood, I., Šabanović, S., Rocha, L. (2014). Designing a Minimalist Socially Aware Robotic Agent for the Home. *Proceedings of the 14th International Conference on the Synthesis and Simulation of Living Systems (Artificial Life 14)*, 876-883.

8. TEACHING

Current Courses

- I453 Computer and Information Ethics
- I341 Prototyping with Arduino Tools
- I400 Explorations in Educational Technology
- I400 Interactive Electronics Studio
- I549 Advanced Prototyping (Graduate level)
- I101 Introduction to Informatics and Computing (New 2025)

Graduate Student Mentoring

- Mentored Yuxing Wu through Luddy PhD Teaching Fellowship (2024)
- Served on two PhD committees (2024)
- Collaborative research mentoring with graduate students Rasika Mulhadrian, Ece Gumel, and Adam Prumm on AI-assisted problem discovery research
- Mentored over 15 graduate students in problem-based design contexts with double-bass players in the Jacobs School and older-adults in Area10 for Aging Co-Design Commons.

9. MAJOR COURSES CREATED/REVISED/ORGANIZED/LED

- **I341 Interactive Electronics Prototyping** (2015-Present) - Developed and taught comprehensive physical computing course from I400 topics course to official course number, teaching hundreds of students hands-on electronics prototyping, 3D modeling, and system design; currently proposing course name change to reflect evolved curriculum beyond Arduino-specific tools
- **I400 Explorations in Educational Technology** (2021-Present) - Created and taught interdisciplinary course combining educational theory with hands-on technology design, taught three times with focus on constructivism, gamification, and physical computing pedagogies

- **I549 Advanced Prototyping** (2024) - Developed graduate-level course integrating physical computing, 3D modeling, and participatory design theory for emerging digital interfaces
- **Luddy Precollege Summer Program Curriculum** (2021-2024) - Led comprehensive curriculum design and implementation for high school STEM education
- **I453 Computer and Information Ethics** – (2014 – 2024) Developed problem-based pedagogy that addresses AI ethics and contemporary technology challenges

10. SERVICE

Administrative Leadership

- Director of Undergraduate Studies, Department of Informatics (2025-Present)
- Director of Curriculum, Luddy Precollege Summer Program (2021-2024)
- Makerspace Director Hiring Committee (2023)